Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 6 [2021 P1]

Test Date: 23rd Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2021-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 10)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 11 – 16)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 23)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 24 – 29)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 30 – 35)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 42)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 43 – 48)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 49 – 54)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

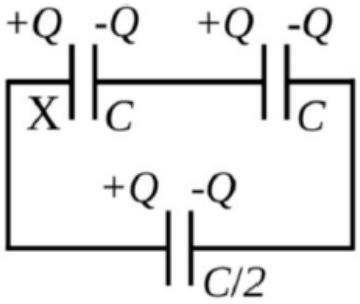
This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases)

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1. An electric circuit consists of three charged capacitors. The capacitances of the capacitors and the values of their charges are given in the diagram. At some moment, the plates of capacitor X are released, allowing them to move freely toward each other. As a result of inelastic collisions, the plates eventually touch. What amount of heat is released in the system as a result?



A) $\frac{Q^2}{2C}$ **B)** $\frac{2Q^2}{3C}$ **C)** $\frac{Q^2}{C}$ **D)** $\frac{2Q^2}{5C}$

2. Match the curve in figure-1 to the curve in figure-2 having the same quality factor in LCR series circuit with AC voltage. Here ϕ is phase difference between current and voltage

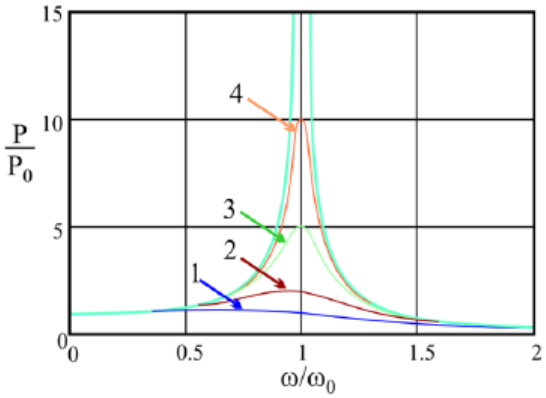


Figure-1

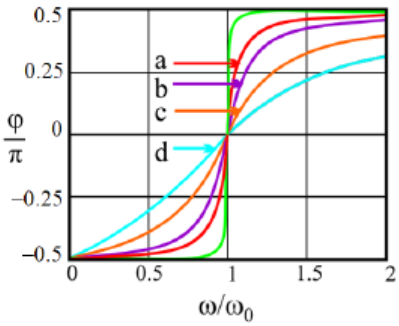


Figure-2

A) 1 → a, 2 → b, 3 → c, 4 → d **B)** 1 → d, 2 → c, 3 → b, 4 → a

C) 1 → a, 2 → b, 3 → d, 4 → c **D)** 1 → d, 2 → c, 3 → c, 4 → d

3. A circular table has radius R and N > 2 equally spaced legs of length h attached to its perimeter. Suppose the table has a uniform mass density with total mass m. Assuming the table does not slip, the minimum horizontal force needed to tip over the table is:-

A) $\frac{mgR}{h}$ **B)** $\frac{mgR}{h}\cos\left(\frac{\pi}{N}\right)$ **C)** $\frac{mgR}{h}\tan\left(\frac{N-2}{2N}\pi\right)$ **D)** $\frac{mgR}{h}\sin\left(\frac{\pi}{2N}\right)$

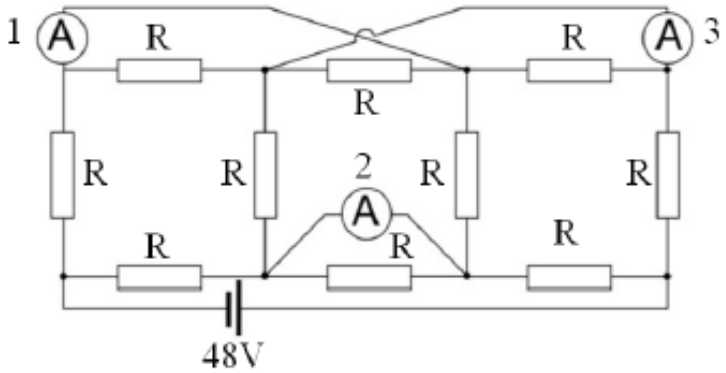
4. The schematic provided has all resistors with a resistance of three ohms, and the battery voltage is 48 volts. All ammeters are ideal.

Statement-1 : Readings of all ammeters is same.

Statement-2 : Reading of ammeter 1 is 5A.

Statement-3 : Reading of ammeter 2 is 7A.

Statement-4 : Reading of ammeter 3 is 7A.



A) All statements are correct.

B) Only statement-1 is correct.

C) Statement-1 & statement-4 are correct.

D) Statement-2 & statement-3 are correct

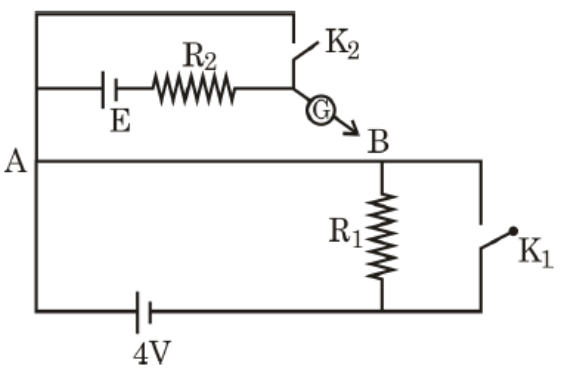
SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Question Stem for Question Nos. 5 and 6

Question Stem

Figure shows the circuit of a potentiometer. The length of the potentiometer wire AB is 50 cm. The emf of the battery is 4 volt, having negligible internal resistance. Values of resistances R_1 and R_2 are 15 ohm and 5 ohm respectively. When both the keys are open, the null point is obtained at a distance of 31.25 cm from end A. Given resistance of wire $AB = 10\Omega$.



5. Potential gradient of wire AB at null point when K_2 is open and K_1 is open is _____ V/cm

6. The balance length when key K_2 is open and K_1 is closed is given by ____ cm.

Question Stem for Question Nos. 7 and 8

Question Stem

The blue of the sky and the red of the sunset are due to a phenomenon called "scattering". When sunlight passes through the earth's atmosphere, much of the light is picked up by the air molecules and given out again in some other direction. Experiments show, in agreement with the theory of scattering, that the shortest waves are scattered more readily then longer waves. To more specific, the scattering is inversely proportional to the fourth power of the wavelength.

$$\text{scattering} \propto \frac{1}{\lambda^4}$$

For every ten violet wave $\left(\lambda = 4\times10^{-7} m\right)$ scattered from a beam, there is only one red wave $\left(\lambda = 7\times10^{-7} m\right)$

Scattering of different colours will be in the ratio as given below :

Violet	Blue	Green	Yellow	Orange	Red
10	6	3	2.5	2	1

7. Wavelength of yellow light will be approximately $\alpha \times 10^{-7}\ m$. Write α.

8. Scattering of wavelength corresponding to threshold wavelength of photoelectric setup of work function 6.2 eV will be nearly η times that violet colour. The value of η is

(Give $hC = 12400\,eV\text{Å}$)

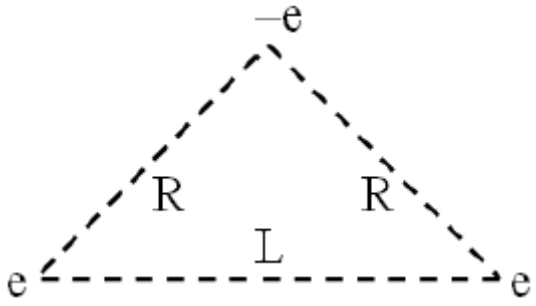
Question Stem for Question Nos. 9 and 10

Question Stem

Two particles with positive charge e and mass M, and a third particle with negative charge – e and mass $m = \frac{M}{2}$, move in circular orbits, maintaining the configuration of an isosceles triangle (see the figure). The planes of the particles' orbits are perpendicular to the base of the triangle.

9. Find the ratio $\frac{R_-}{R_+}$ where R_- is the radius of the orbit of the negatively charged particle, and R_+ is the radius of the orbit of the positively charged particles. Round your answer to the nearest whole number.

10. Find the ratio of the base length L of the triangle to the length of its side R. Round your answer to two decimal places.



SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

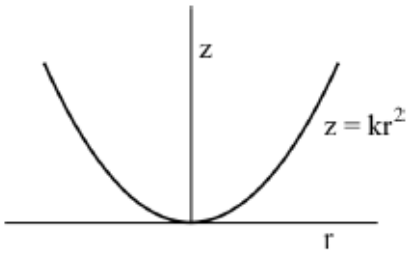
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

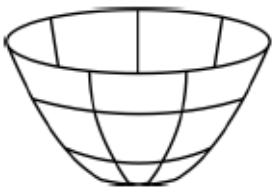
Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and

Choosing any other option(s) will get -2 marks.

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11. A particle is constrained to move on the inner surface of a frictionless parabolic bowl whose cross-section has equation $z = kr^2$. The particle begins at a height z_0 above the bottom of the bowl with a horizontal velocity v_0 along the surface of the bowl. The acceleration due to gravity is g.





A) The value of v_0 for which the particle moves in a horizontal circle is $\sqrt{gz_0}$

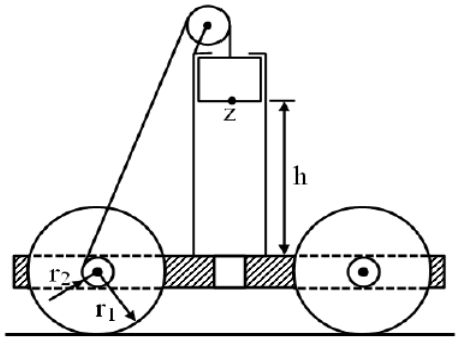
B) For $v_0 > \sqrt{2gz_0}$ **.** The the maximum height reached by the particle is $\dfrac{v_0^2}{2g}$

C) If $v_0 = 0$ and z_0 is very small, then the time in which the particle returns to its nitial point is $4\sqrt{\dfrac{2z_0}{g}}$

D) If $v_0 = 0$ and z_0 is very small, then the time in which the particle returns to its initial point is $\dfrac{\pi\sqrt{2}}{\sqrt{kg}}$

12. A cart for studying uniformly accelerated motion is driven by a weight, as shown in Fig. A descending weight of mass m rotates a rear axle via a rope passing over a pulley, with the rope sufficiently wound around the axle. Initially, the weight is at a height h above the cart. When released, the cart starts moving. Each of the four wheels of the cart has a mass m_1 and a radius r_1. The rear and front axles have a mass m_2 and a radius r_2. The wheels and axles are made as solid homogeneous cylinders. The motion of the weight is guided by a smooth vertical tube. The mass of the cart without the weight, axles, and wheels is

negligible. The mass of the pulley, friction of the weight inside the tube, and rolling resistance of the wheels are considered negligible.



A) The acceleration of the cart is $\dfrac{mgr_2}{T_1\left[m\left(\dfrac{r_2^2}{r_1^2}\right)+6m_1+m_2\left(2+\dfrac{r_2^2}{f_1^2}\right)\right]}$

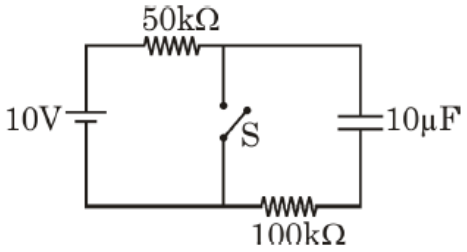
B) The speed of the cart till the weight falls on the cart is

$$\sqrt{\frac{2mgh}{m\left(1+\dfrac{r_2^2}{r_1^2}\right)+6m_1+m_2\left(2+\dfrac{r_2^2}{f_1^2}\right)}}$$

C) The tension in the string while the weight is falling is $m\left(g-a\dfrac{r_2}{r_1}\right)$, where a is the acceleration of the cart.

D) The tension in the string while the weight is falling is $m\left(g-a\dfrac{r_2^2}{r_1^2}\right)$, where a is the acceleration of the cart.

13. In the circuit shown, the switch S has been closed for a long time. At time t = 0, the switch is opened. It remains open for a time T, after which it is closed again.



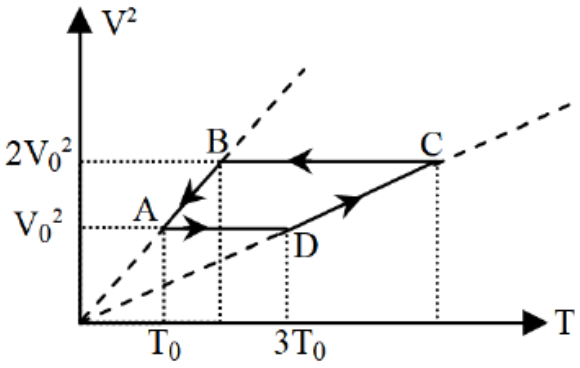
A) Voltage drop across $100k\Omega$ resister is $10e^{-t/1.5}V$ *for* $t<T$.

B) Voltage drop across $100k\Omega$ resister is $\dfrac{20}{3}e^{-t/1.5}V$ for t<T

C) Voltage drop across $100k\Omega$ resister is $10\left(e^{T-t}-e^{(T-3t)/3}\right)V$ for t<T

D) The time constant of the circuit is 1.5 sec for t > T.

14. One mole of ideal monoatomic gas follows cyclic process as shown in the figure. Choose the correct option(s):



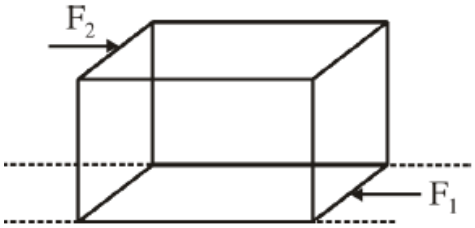
A) $\frac{T_B + T_C}{2} = 4T_0$

B) Molar heat capacity for process B → A is 2R

C) Molar heat capacity for process D → C is 4R

D) Work done by gas during D to C is $\frac{3RT_0}{2}$

15. A cubical container has 2 fixed vertical walls and two vertical walls can freely rotate about one of the horizontal edges. The right wall can rotate about upper edge and the left wall can rotate about the bottom edge. Water is filled upto the top and forces F_1 and F_2 (perpendicular to the walls) are applied to keep the walls at from opening as shown in figure. Given that the cube is at rest on a rough horizontal surface. (Ignore atmospheric pressure)



A) Friction on the box from the surfaces is to the left

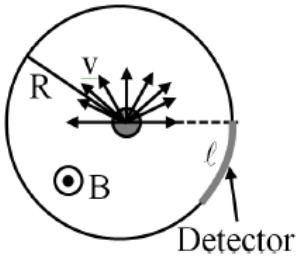
B) Friction on the box from the surfaces is to the right.

C) Normal reaction is shifted towards left wall.

D) Normal reaction passes through the center.

16. At the center of a tube with radius R, a source emits charged particles (see diagram). Each particle has a charge q > 0 and mass m. All particles are emitted from the source in the plane of the drawing with the same speed v, evenly distributed in the upper half of the tube. A detector of length ℓ, which registers the total number of particles that reach it in one unit of time, is fixed on the surface of the tube as shown in the diagram. Along the entire tube (in the direction "toward us"), a uniform magnetic field of induction B is applied. For what values of B will the detector readings be at their maximum possible

values? The interaction between charged particles and radiation is negligible. [Take $\ell = \frac{\pi R}{6}$]



A) $\frac{mv}{qR}$ **B)** $\frac{3}{2}\frac{mv}{qR}$ **C)** $\frac{2mv}{qR}$ **D)** $\frac{5}{2}\frac{mv}{qR}$

SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** ***: +4*** If ONLY the correct integer is entered;
- ***Zero Marks*** ***: 0*** In all other cases.

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17. A steel girder is planted securely between two sides of a ravine in order to provide a bridge. The total cross-sectional area of the girder is $30cm^2$, and the length of the girder is 4.0 m. Installed at a temperature of $5°C$, the temperature now rises to $20\ °C$. Calculate the force exerted (in N) by the girder due to the change in temperature, assuming the ends do not move. Young modulus of steel $= 2.0\times10^{11}$ Pa. Coefficient of linear expansion is $1.2\times10^{-7}\ °C^{-1}$. If the force is 216P, the value of P is

18. A closed cylindrical soft drink can of height 12cm and radius 4cm is filled up to height 9cm. Volume above liquid has air at a pressure 10 k Pa. Soft drink is incompressible and has specific gravity 0.8. Taken far away from all celestial bodies in a spaceship where can is rotated about its axis with constant angular velocity $\frac{250}{\sqrt{3}}$ rad/s , it exerts pressure $(10000\times N)$ Pascal on curved surface of can. N is ……

19. An ornament is lying at the bed of a pond of depth $8\sqrt{3}m$**.** Pond is full of water whose refractive index is $\sqrt{\frac{3}{2}}$ A diver at the surface of pond is trying to look at ornament such that her line of vision makes 30° with the surface of pond. Perpendicular distance of image of ornament from surface of water in metres is.....

CHEMISTRY

Max. Marks: 60

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

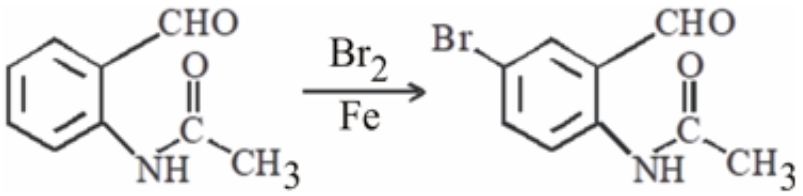
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Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

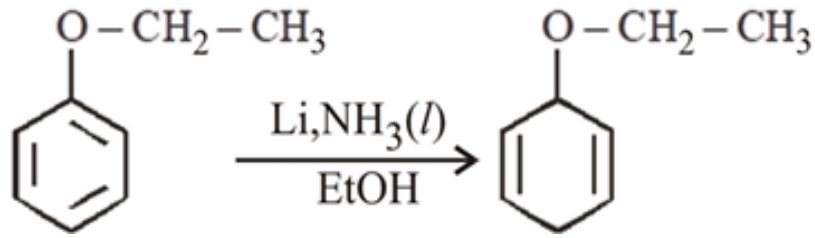
- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases.

20. Which of the following reaction represents the correct major product :

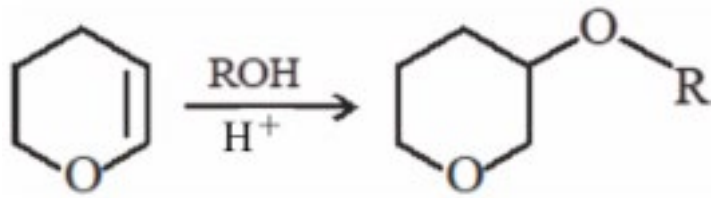
A)



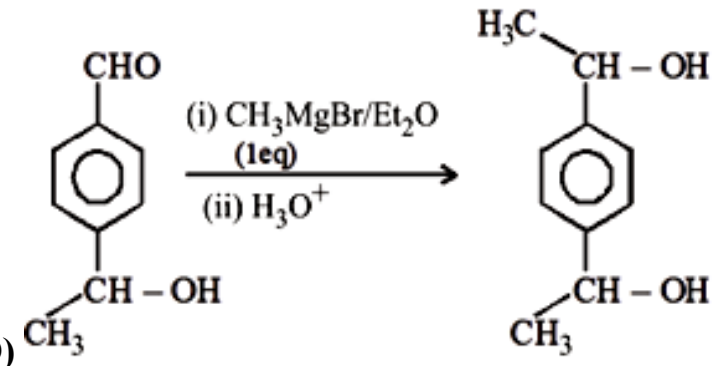
B)



C)



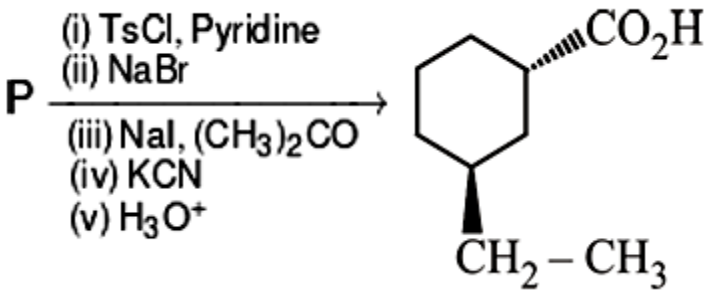
D)

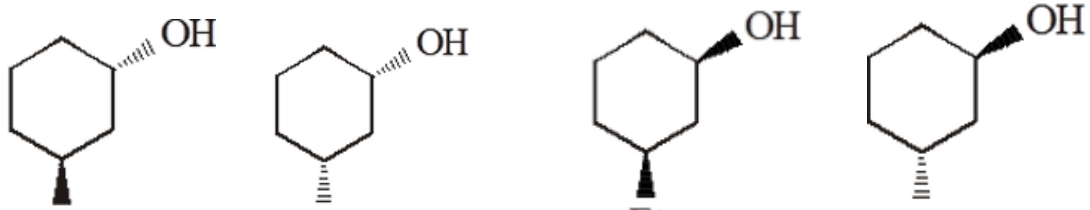


21. A solution of weak acid HA was titrated with strong base KOH. The weak acid is completely neutralized by adding 60 mL of 0.1 M KOH. Now, 20mL of 0.1 M HCl was added to titrated solution, the pH was found to be 5.3. What will be the pH of solution obtained by mixing 50 mL of 0.1 M KOH and 50 mL of 0.1 M HA? (log 2= 0.3)

A) 8.00 **B)** 8.7 **C)** 8.85 **D)** 9.00

22. In the given reaction sequence the correct compound P is:





A) **B)** **C)** **D)**

23. Which of the following statement is correct?

A) $\left[CoF_6\right]^{3-}$ is d^2sp^3 hybridized and diamagnetic in nature

B) $\left[NiCl_4\right]^{2-}$ is sp^3 hybridized and diamagnetic in nature

C) $\left[Ni(CO)_4\right]$ is dsp^2 hybridized and diamagnetic in nature

D) $\left[Cu(NH_3)_4\right]^{2+}$ is dsp^2 hybridized and paramagnetic in nature.

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

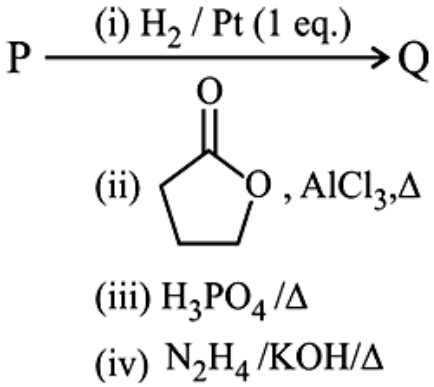
Question Stem for Question Nos. 24 and 25

Question Stem

A compound $P(C_9H_{10})$ gives benzaldehyde and aliphatic aldehyde on reductive ozonolysis. This aliphatic aldehyde gives positive iodoform test. Identify P and answer the given questions.

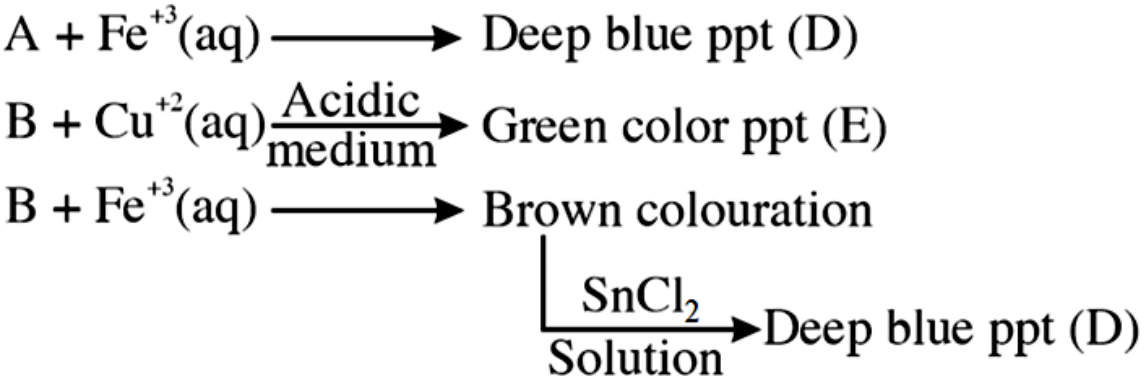
24. What is the total number of major organic products formed [through most stable carbocation] when P is reacted with HBr / CCl_4 ?

25. Find degree of unsaturation of the product Q?



Question Stem for Question Nos. 26 and 27

Question Stem



26. The number of unpaired electrons in compound 'B' is **'p'** and number of atoms in a molecule of compound E are **'q'**, then $|p - q|$ is:

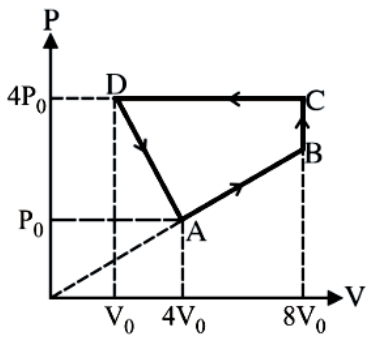
27. When 'A' is treated with conc. H_2SO_4, a gas is produced which is colorless, odourless and poisonous. The molar mass of gas (g/mol) is 'x' and oxidation state of central metal in compound 'A' is 'y'. Then, value of $|x + y|$ is :

Question Stem for Question Nos. 28 and 29

Question Stem

0.06 moles of an ideal monoatomic gas undergoes a process as shown in figure. The temperature at point 'A' is 100 K.

$\left(\text{Use}: R = \frac{25}{3} J / \text{mol.K}, 1\text{Lbar} = 100J\right)$



28. The magnitude of work done (in Joule) in cyclic process ABCD is

29. The magnitude of enthalpy change (in L.bar) in process CD is:

SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

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30. Among the following, the correct statement(s) is/are:

A) The two six-membered cyclic hemiacetal forms of D-(+)-glucose are called as C2 epimers.

B) Invert sugar is prepared by acid catalyzed hydrolysis of maltose.

C) DNA contains cytosine and thymine as pyrimidine bases.

D) Glycine is an optically inactive amino acid.

31. Which of the following option is/are correct for ionic solids?

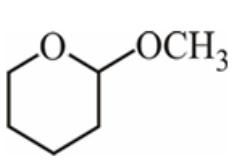
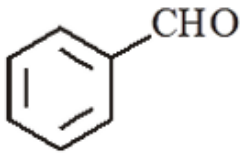
A) In ZnS (Zinc blende) structure, distance between two nearest Zn^{+2} ions is $5\,\text{Å}$, then distance between two nearest S^{2-} ions is $0.707\,\text{Å}$.

B) In NaCl (rock salt) structure, the second nearest cation for any anion placed at corner is present at a distance of $\frac{\sqrt{3}}{2}a$ (where 'a' = edge length of unit cell) and number of second nearest cations are 8.

C) In NaCl (rock salt) structure, the third nearest anion for any anion placed at corner is present at a distance of $\frac{\sqrt{3}}{2}a$ (where 'a' = edge length of unit cell).

D) The packing fraction of CsCl (8 : 8) structure is $\frac{\sqrt{3}\pi}{4}$ (approx.) (Given that: $r_{Cs^+} = 170\text{pm}, r_{Cl^-} = 180\text{pm}$)

32. Which of the following compound(s) will not give silver mirror on reaction with ammoniacal silver nitrate solution?

A) formic acid **B)** Propane **C)**  **D)** 

33. 2.84 g of mixture of $CaCO_3$ and $MgCO_3$ was dissolved in 250 mL of 0.4 M HCl(excess) solution which was then diluted to 400 mL. If 10 mL of resulting solution requires 10 mL of $\frac{M}{20} Na_2CO_3$ for neutralization. Then,

A) The mass percentage of $CaCO_3$ in original mixture is 70.42 % (approx.)

B) The mole percentage of $MgCO_3$ in original mixture is 33.33% (approx.)

C) The number of equivalents of HCl used with original mixture was 0.06.

D) The number of mole of HCl neutralized by Na_2CO_3 is 0.01.

34. The incorrect statement(s) related to metallurgical processes is (are)

A) In leaching process of aluminium ore (Bayer process), a small amount of freshly prepared sample of hydrated Al_2O_3 is added to induce precipitation.

B) Wrought iron is the purest form of commercial iron and is prepared from cast iron by reducing elemental impurities in a furnace lined with haematite

C) Copper matte primarily consists of Cu_2O and a smaller amount of FeS.

D) Extraction of zinc from zinc blende involves calcination followed by reduction with carbon.

35. The incorrect statement(s) related to colloids is/are.

A) The colloid ion is a 4% solution of nitro-cellulose in a mixture of alcohol and ester

B) A mixture of milk and water appears red when viewed by the reflected light and blue when viewed by the transmitted light.

C) Sols of starch, gum, clay and methylene blue are negatively charged colloidal sol

D) When the ionic strength of a colloidal solution is increased, thickness of double layer is decreased and the colloid gets precipitated.

SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** *: +4* If ONLY the correct integer is entered;
- ***Zero Marks*** *: 0* In all other cases.

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36. In the following reaction sequence, the amount of T (in g) formed from 12 mole CaC_2 is___

[Atomic weight in $gmol^{-1}$: H = 1, C = 12, N = 14, O = 16, Ca = 40.] The yield (%) corresponding to the product in each step is given in the parenthesis

$$CaC_2 \xrightarrow{H_2O} \underset{(100\%)}{P} \xrightarrow[\text{iron tube}]{\text{red hot}} \underset{(60\%)}{Q} \xrightarrow[\text{conc. } H_2SO_4]{\text{conc. } HNO_3,} \underset{(50\%)}{R} \xrightarrow{Sn/HCl} \underset{(50\%)}{S} \xrightarrow{Ac_2O} \underset{(100\%)}{T} + CH_3COOH$$

37. A sample of excited single electron ions can emit maximum 10 different spectral lines when de-excited to ground level. The maximum energy emitted during transition is 117.5 eV. Now, photons of minimum energy emitted during transition strikes on a metal having work function 1.254 eV emits photoelectrons. The de-Broglie wavelength of emitted photo electron (in Å) is:

38. A sample of water is electrolysed during an experiment using 40 F electricity with 75% current efficiency. The released O_2 is used in a bomb calorimeter to burn 3.5 g C_2H_4 gas at 300 K. The temperature of the calorimeter was found to increase from 300 K to 300.50 K due to combustion process. The magnitude of heat of combustion at constant pressure (in kJ) for 5 moles of ethene is: (Given: Heat capacity of the calorimeter is 3 kJ/K, R = 8 J/mol.K)

MATHEMATICS

Max. Marks: 60

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)

Negative Marks: -1 (In all other cases

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39. Straight line $L:\ 2x\ +\ y\ =\ 4$ is tangent to an ellipse whose foci are $(1,\ 3)$ & $(2,\ 4)$. Straight line L_1 parallel to line L also touches this ellipse. Point of contact of L_1 to this ellipse is

A) $\left(\frac{53}{14},\frac{65}{14}\right)$ **B)** $\left(\frac{61}{25},\frac{103}{25}\right)$ **C)** $\left(\frac{15}{7},\frac{31}{25}\right)$ **D)** $\left(\frac{63}{25},\frac{103}{25}\right)$

40. $\int_1^2\left(2^x+2^{-x}\right)d\left(2^x\right)=$

A) $\frac{5}{4}+ln\,2$ **B)** $\frac{3}{2}+\ln 2$ **C)** $\frac{3}{2}-ln2$ **D)** $\frac{7}{8}+ln2$

41. $\lim\limits_{x\to 0}\dfrac{xe^{sinx}-e^x sin^{-1}(sinx)}{sin^2x-xsinx}=$

A) -2 **B)** -1 **C)** 1 **D)** 2

42. If the standard deviation of 20 observations $x_1,x_2,x_3,.....,x_{20}$ is 4 and that of another 20 observations $y_1,y_2,y_3,.....,y_{20}$ is 3. Also $X_i=\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)$ where $\bar{x}$ *and* $\bar{y}$ are means of x_i' and y_i' respectively. If $\sum_{i=1}^{20}X_i=90,$ then the standard deviation of the observations $x_1-y_1,x_2-y_2,x_3-y_3,.....,x_{20}-y_{20}$ is divisible by

A) 5 **B)** 4 **C)** 3 **D)** 7

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

Question Stem for Question Nos. 43 and 44

Question Stem

Let a, b and c be real numbers such that the system of linear equations

$10x + 11y + 12z = a$

$13x + 14y + 15z = b$

$16x + 17y + 18z = c - 3$

is consistent. Let $|x|$ denotes the determinant of matrix

$$X = \begin{bmatrix} a & 2 & 1 \\ b & 1 & 0 \\ c & 0 & -1 \end{bmatrix}$$

Let P be the plane containing all those (a, b, c) for which system of linear equations is consistent and D be the square of the distance of the point $(-9, 0, 0)$ from the plane P.

43. The value of $(|x| + 10)$ is _____

44. The value of D is _____

Question Stem for Question Nos. 45 and 46

Question Stem

Let f be a polynomial function defined only over non negative real numbers such that

$f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ and f(x) is one-one function with

$f(0) = 1$ *and* $f'(1) = 2$ then

45. The value of f(5) is

46. If $h(x) = \min\left\{\frac{2}{f(x)}, x^2, |1 - |x||\right\}$, where $x \geq 0$, then the number of points of non differentiability of h(x) is

Question Stem for Question Nos. 47 and 48

Question Stem

Let S_n be the sum of first n terms of the sequence $\{a_n\}$. Let $a_n = 5S_n + 1$ holds for any $n \epsilon N$.

Let $b_n = \frac{4 + a_n}{1 - a_n}$, $n \in N$

47. $\left(\frac{4^{10}a_{10}+5}{2}\right)=$

48. $\left(\frac{\left(4^{12}-1\right)b_{12}-1}{2^{27}}\right)=$

SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** **: +4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks*** **: +3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks*** **: +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks*** **: +1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks*** **: 0** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

49.

$$\int\frac{x^4+32}{x^8(x^4-16)}dx=\frac{3}{\lambda}ln\left|\frac{x-2}{x+2}\right|-\left|-\frac{\mu}{256}\tan^{-1}\frac{x}{2}+\frac{1}{\alpha x^3}+\frac{2}{7x^7}+c\right.$$

Then,

A) $\lambda = 512$ **B)** $\mu = 5$ **C)** $\mu = 3$ **D)** $\alpha = 15$

50. Let $A(3,\ 0,\ 4)$ *and* $B(4,\ 6,\ 0)$ be two given points and $P(\alpha,\ 0,\ 0)$ be point on x-axis such that $PA + PB$ is minimum, then

A) If point Q is on the y-axis such that $QA+QB$ is minimum then area of ΔPOQ is $\frac{21}{4}\ (unit)^2$ where O is origin

B) The volume of tetrahedron $OPAB$ (where O is origin) is $\frac{68}{5}\ (unit)^3$

C) Perpendicular distance from the point A to the plane containing the points P, Q, B is 4 (units)

D) If point Q is on the y-axis such that $QA+QB$ is minimum then area of ΔPOQ is $\frac{17}{3}(unit)^2$ where O is origin

51. If $\left(f(x)-2\right)\left(x^4+x^2+1\right)-\left(f(x)+2\right)\left(x^2+x+1\right)^2=0$ is true for all $x\epsilon R-\{0\}$, then which of the following statement(s) is/are true

A) f(x) has a local maxima at x = –1 **B)** $\left|f(x)\right| \geqslant 4, \forall x \in R-\{0\}$

C) f(x) has a local maxima at x = 1 **D)** The value of $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (secx) f(x)dx = 0$

52. If P_n denotes the number of ways in which n runners participating in a race can reach the finishing line, when one or more runners can reach at same time, then.

A) $P_4 = 74$ **B)** $P_4 = 75$ **C)** $P_5 = 541$ **D)** $P_5 = 521$

53. Let roots of quadratic equation $8x^2 - 14x + 3 = 0$ are α and β where $\alpha < \beta$. Which of the following equation has $\left(2\alpha + \sqrt{\frac{\beta}{2}}\, i\right)^{50}$ *and* $\left(2\alpha - \sqrt{\frac{\beta}{2}}\, i\right)^{50}$ as roots :

A) $x^2+x+1=0$ **B)** $x^3+1=0$ **C)** $x^3-1=0$ **D)** $x^2-x+1=0$

54. If $\ln x \leq \lambda x^3,\ \forall x \in (0,\infty)$ then possible value/s of λ is are

A) $\frac{1}{5}$ **B)** $\frac{1}{10}$ **C) 1** **D)** $\frac{1}{e^2}$

SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** ***: +4*** If ONLY the correct integer is entered;
- ***Zero Marks*** ***: 0*** In all other cases.

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55. A ship sends five different codes A, B, C, D, E for communication, but each week uses only one of them. The code used in a definite week is randomly selected with equal chance among the four ones that have not been used in the last week. Suppose the code used in the first week is A. If the probability that A is also used in the ninth week is $\frac{\alpha}{\beta}$ (where α, β are relatively prime positive integers) then the value of $\left[\frac{\alpha}{100}\right]$ = (where [.] is greatest integer function)

56. If z is a complex number such that $|z| = 1$. If maximum value of $\left|z^3 - z + 2\right|$ is 'a', then the sum of digit of a^2 is.

57. If the area bounded by $x = 0, y = 0, x = 2, y = 2, y \leqslant e^x$ *and* $y \geqslant \ln x$ is $(a - b\ln 2)$ sq. unit, where $a \in N,\ b \in N$ then value of $\left(\frac{a+b}{2}\right)$ is